# INFINITE TUTORIAL - OFFICIAL EXAMINATION PAPER

## CBSE - CLASS 10 MATHEMATICS (REAL NUMBERS & POLYNOMIALS)

Max Marks: 50 | Duration: 90 Minutes | Subject Code: MAT10

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Candidate Name: _____ Roll Number: _____ Date: _____

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General Instructions: All questions are compulsory. Show detailed intermediate steps.

SECTION A - OBJECTIVE QUESTIONS (1 Mark Each)

Q1. If two positive integers a and b are written as a = x^3 y^2 and b = x y^3, find HCF(a, b).

Q2. The total number of zeroes of a quadratic polynomial cannot exceed: 2.

Q3. If alpha and beta are zeroes of x^2 - 5x + 6, find the value of alpha + beta.

SECTION B - SHORT ANSWER (2 Marks Each)

Q4. Prove that 3 + 2*sqrt(5) is irrational given that sqrt(5) is irrational.

Q5. Find a quadratic polynomial whose sum and product of zeroes are -3 and 2.

SECTION C - LONG QUESTIONS (3 Marks Each)

Q6. Using Euclid Division Lemma, find the HCF of 4052 and 12576.

Q7. Find the zeroes of polynomial 4s^2 - 4s + 1 and verify relationship with coefficients.

SECTION D - ADVANCED PROBLEMS (4 Marks Each)

Q8. If one zero of 2x^2 - 3x + k is reciprocal of the other, find the value of k.

Q9. An army contingent of 616 members is to march behind an army band of 32 members in a parade. Find maximum columns.

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*** End of Question Paper - Certified by Infinite Tutorial Academic Directorate ***